ThreatShield AI

Real-time ransomware detection with explainable risk scoring and automated data protection

Domain

Domain 2: AI Data, Storage & Security. Problem 2D: Real-Time Threat Detection to Prevent Breaches & Ransomware

Problem

Ransomware can encrypt thousands of files within minutes. Signature-based tools miss new variants, and manual response is too slow, so data is lost before anyone reacts. Small organizations without a security team are hit hardest.

Target users

IT administrators and security analysts in SMEs, schools and clinics, and owners of small organizations with no dedicated security team.

Team: The Phenomenals

Prince Jain	Team lead, backend and API
Sanidhya Talwai	ML detection engine
Anvit Naik	Endpoint agent and response
Richa Gupta	Dashboard and alert UI
Aishwarya Raikar	Simulator, testing, metrics

Score behavior, explain it, contain it, protect the data

A lightweight agent watches endpoint behavior, scores each process from 0 to 100, and responds in tiers before encryption spreads.

Core features

- Monitors process, file and network activity
- Risk score from five signals: write rate, renames and extension changes, write entropy, canary-file access, process behavior
- Isolation Forest plus rule engine
- Alert shows the reasons behind the score

Objectives

- Detect ransomware-like behavior within 5 seconds
- Keep false positives under 10%
- Limit the number of files lost
- Make every alert explainable

Intended benefits

- Fewer files lost and faster recovery
- Alerts non-experts can act on
- No signature updates or labeled attack data needed
- Affordable for SMEs, schools and clinics

Response tiers Score 40-70: alert | 70-90: suspend process and snapshot files | 90+: terminate process

Beyond signatures: lightweight, explainable, data-first

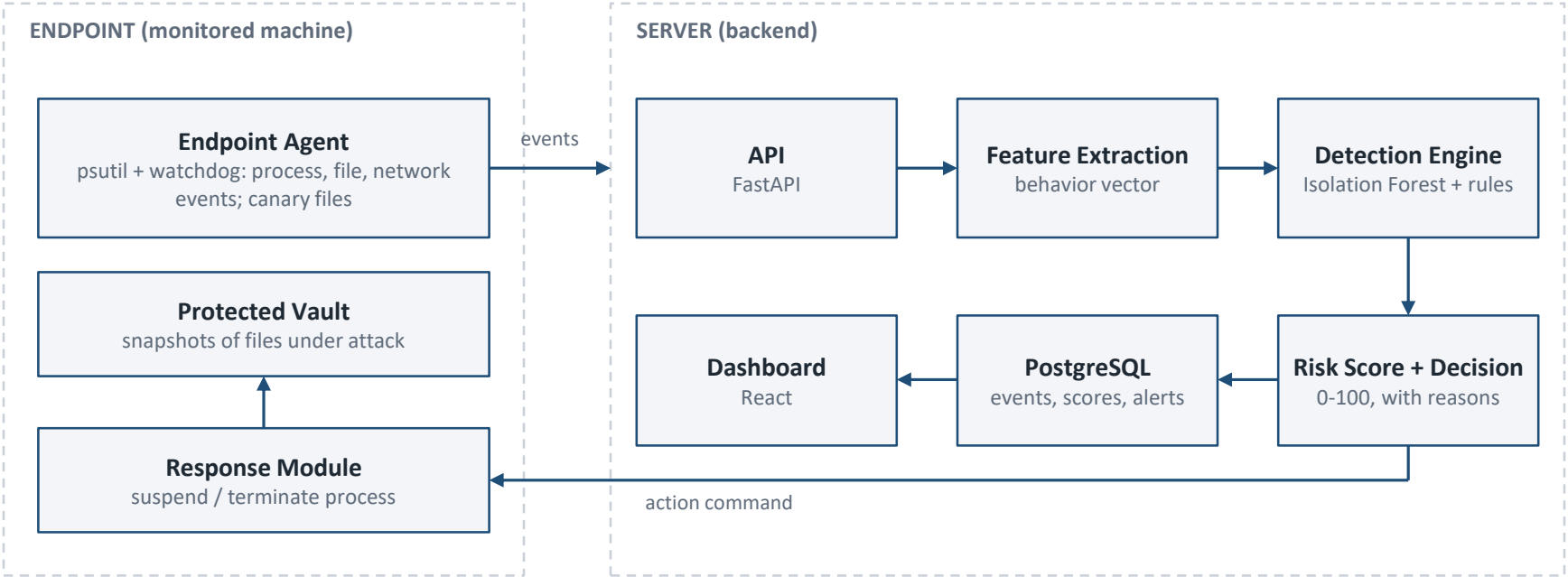
Existing approach	Limitation	ThreatShield AI
Signature antivirus	Misses new, fileless and zero-day threats	Detects by behavior; no signature needed
Enterprise EDR/XDR (Defender for Endpoint, CrowdStrike, SentinelOne)	Strong, but costly and complex; usually needs a security team; alerts are hard for non-experts to read	Open-source, lightweight, SME-friendly; plain-language reasons on every alert
Backup and recovery tools	Restore data after the damage; do not stop the attack	Snapshots files under attack and stops the process
Manual SOC monitoring	Too slow for fast encryption	Automatic tiered response within seconds

What makes it different

- Explainable score: each alert lists the signals that raised it, such as files modified per second, high write entropy, or a canary file touched
- Data-first response: files under attack are copied to a protected vault before the process is terminated; canary files trigger an instant high score
- Tiered response: alert, then suspend, then terminate, so a false positive does not kill a legitimate process

Behavior-based detection itself is not new. Our contribution is the explainable, data-protecting, low-cost combination.

System architecture and data flow



User workflow 1. Analyst opens the dashboard 2. Sees the alert with its reasons 3. Reviews the contained process and restores files from the vault

Open-source stack, five owners, 24-hour build plan

Technology and why

Python	One language for agent and ML
psutil, watchdog	Simple process and file monitoring; no Sysmon setup
scikit-learn Isolation Forest	Unsupervised; needs no labeled ransomware data
FastAPI	Fast to build, async event intake
PostgreSQL	Reliable store for events and alerts
React	Live dashboard for alerts and timeline

Team responsibilities

- Prince: API, database, integration
- Sanidhya: Features, model, scoring
- Anvit: Agent, response, vault
- Richa: Dashboard, alert view
- Aishwarya: Simulator, tests, metrics

Timeline (24 hours)

0-3 h	3-8 h	8-13 h	13-17 h	17-21 h	21-24 h
Setup, agent telemetry	Simulator, features, baseline data	Detection and risk score	Response, vault, canary files	Dashboard, integration	Evaluation, demo rehearsal

A demo we can build, measure and scope honestly

Hackathon deliverables

- Agent, detection engine, risk score
- Tiered response with file snapshots
- Live dashboard with alert reasons
- Ransomware simulator for testing

Demo plan

- Simulator encrypts files in a sandbox folder
- Dashboard shows score rise and reasons
- Process is terminated, files restored from vault
- Benign backup and zip runs raise no kill

Measures of success

- Detection rate 90% or higher over 20 simulated runs
- False positive rate under 10% over 20 benign workloads
- Detection under 5 s from first malicious write
- Files lost before containment, reported per run

Future enhancements

- Sysmon / Windows event log input
- MITRE ATT&CK mapping
- Network isolation of the host
- Multi-endpoint fleet management

Risks and dependencies

- Benign tools that mimic ransomware: allowlist, entropy checks, alert-only tier
- Limited training data: synthetic benign activity plus our simulator
- Setup time: fallback is monitored-folder mode

Resources required

- Datasets: synthetic only, generated by us
- Equipment: laptops, one Windows VM
- Paid services: none; all open-source
- External access: none; AI coding assistants used for development support